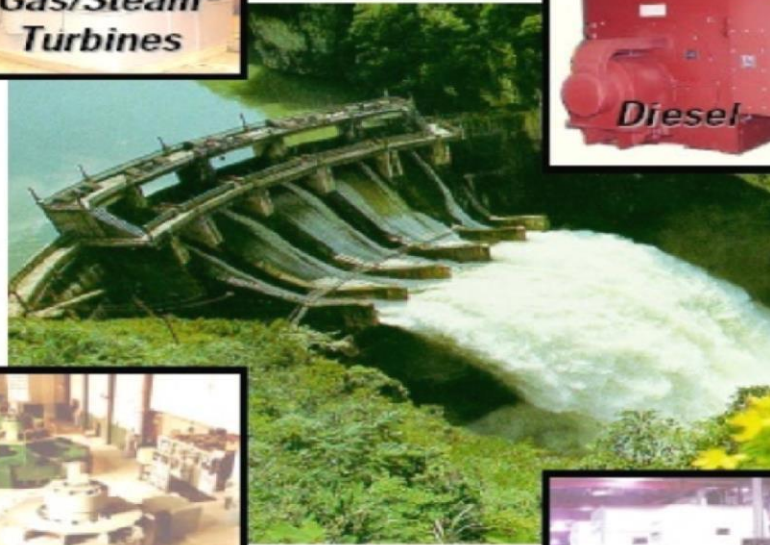
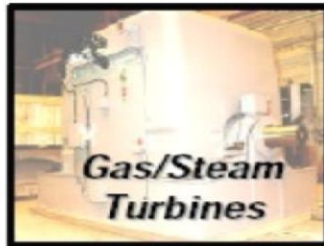
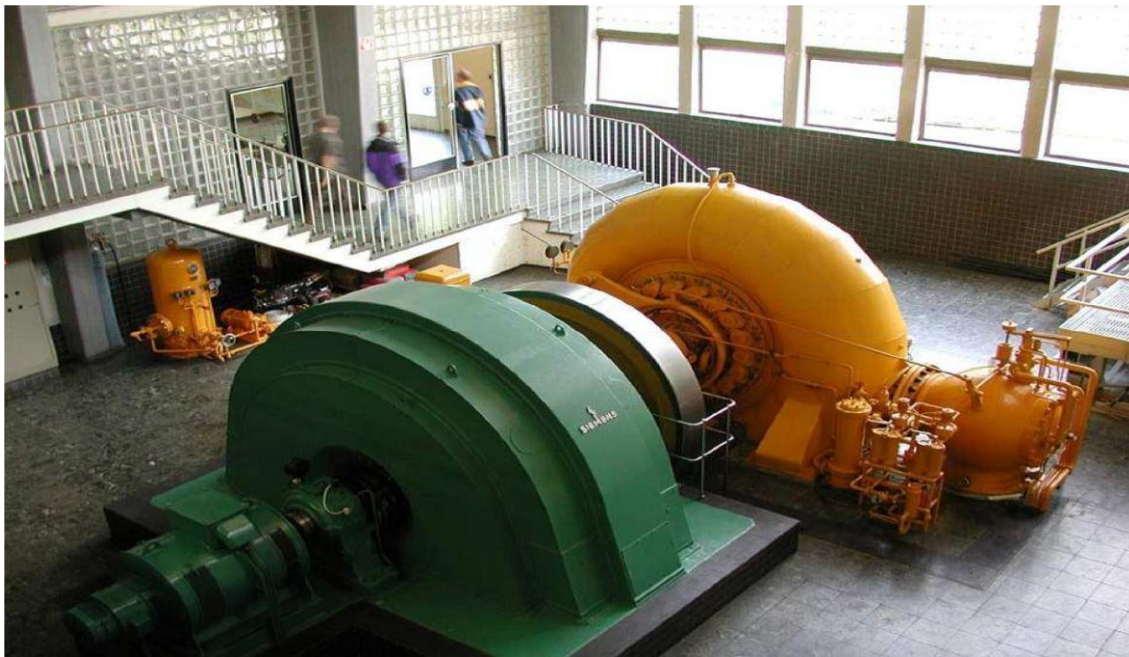




Synchronous Machines

Synchronous Machines

- *Synchronous generators* or *alternators* are used to convert mechanical power derived from steam, gas, or hydraulic turbine to ac electric power
- Synchronous generators are the primary source of electrical energy we consume today
- Large ac power networks rely almost exclusively on synchronous generators
- *Synchronous motors* are built in large units compare to induction motors (Induction motors are cheaper for smaller ratings) and used for constant speed industrial drives

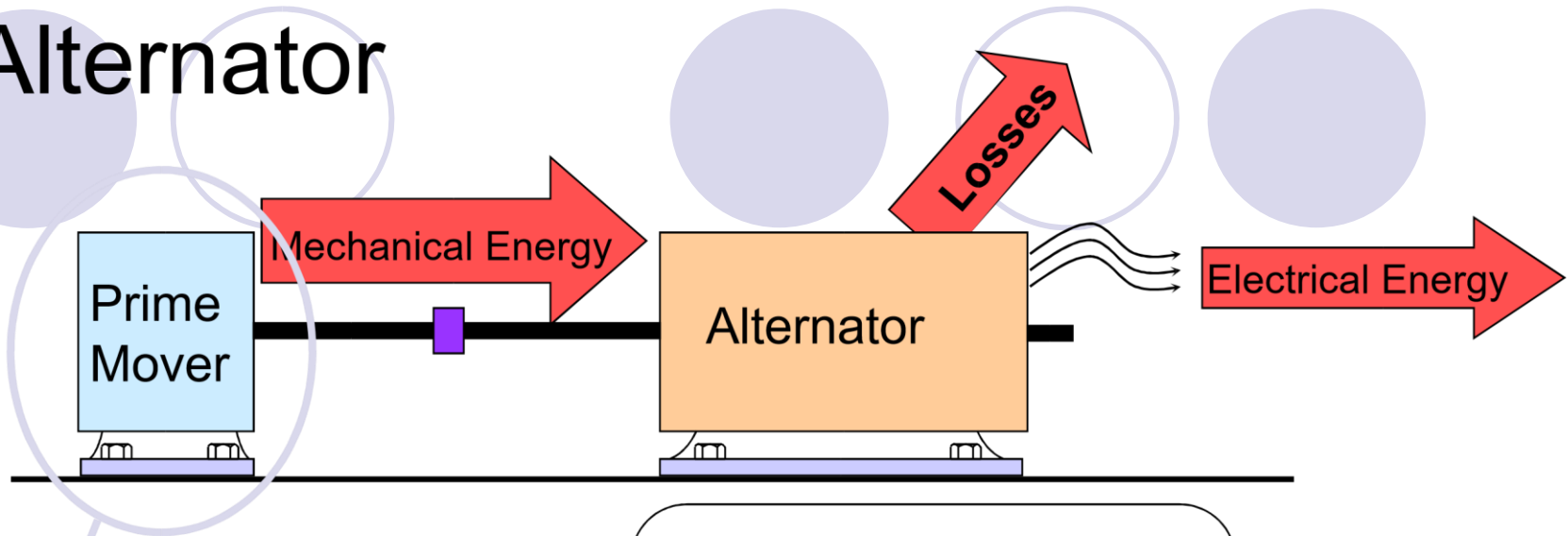




KW Genset / Generator Set, powered with a

3 phase Dunlite machine

Alternator



- Steam Turbine
- Small petrol engine
- Diesel Engine

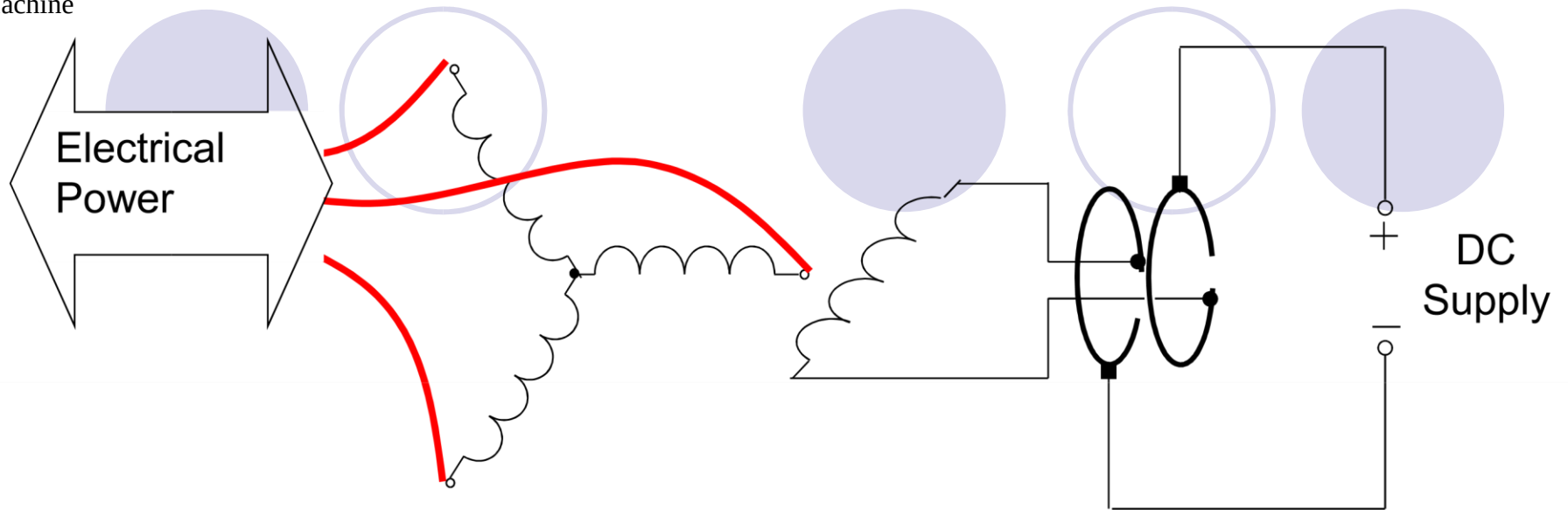
Alternator:

$$P_{in} = P_{out} + \text{Losses}$$

Alternator:

$$\text{Eff}\% = \frac{P_{out}}{P_{in}} \times 100$$

Synchronous Machine



Stator

- Identically wound to an induction motor.
- Connected to supply.

Rotor

- Constant DC field
- Connected to supply via sliprings.

Construction

- A DC current is applied to the rotor winding, which then produces a rotor magnetic field. The rotor is then turned by a prime mover (eg. Steam, water etc.) producing a rotating magnetic field.
- This rotating magnetic field induces a 3-phase set of voltages within the stator windings of the generator.
- “Field windings” applies to the windings that produce the main magnetic field in a machine, and □ “Armature windings” applies to the windings where the main voltage is induced.
- For synchronous machines, the field windings are on the rotor, so the terms “rotor windings” and “field windings” are used interchangeably.

Construction

Ø Basic parts of a synchronous generator:

- **Rotor - dc excited winding**
- **Stator - 3-phase winding in which the ac emf is generated.**
- **The manner in which the active parts of a synchronous machine are cooled determines its overall physical size and structure**
- **Stator of synchronous machines is the same as the stator of induction machines.**

Types of Rotor of synchronous Machines

The rotor of a synchronous generator is a large electromagnet and the magnetic poles on the rotor can either be salient or non salient construction. Non-salient pole rotors are normally used for rotors with 2 or 4 poles rotor (**High Speed**), while salient pole rotors are used for 4 or more poles rotor (**Low Speed**).

q Salient-pole synchronous machine q Non Salient Cylindrical or round-rotor synchronous machine **Salient Pole**

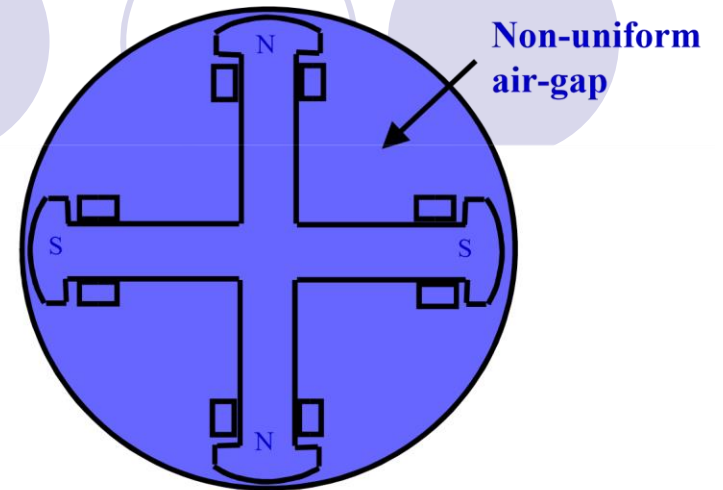
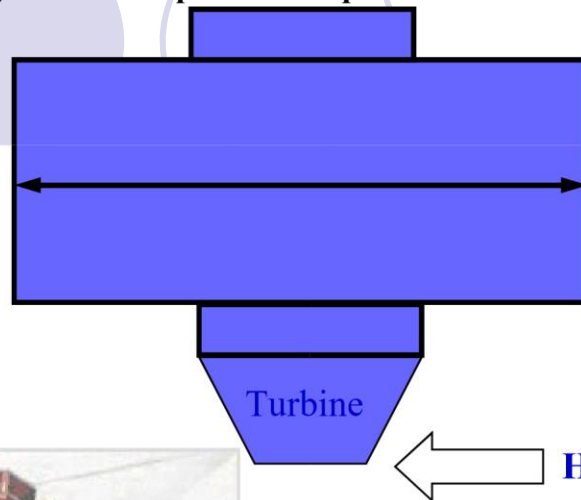
Rotors

- 1.** The salient rotor has protruding poles.
- 2.** The salient rotor has several poles.
- 3.** The salient rotor has large diameter (2 to 6m) to accommodate several poles and short axial length (1m).
- 4.** Salient pole rotor are used for low speed Hydro generators.

Salient-Pole Synchronous Generator

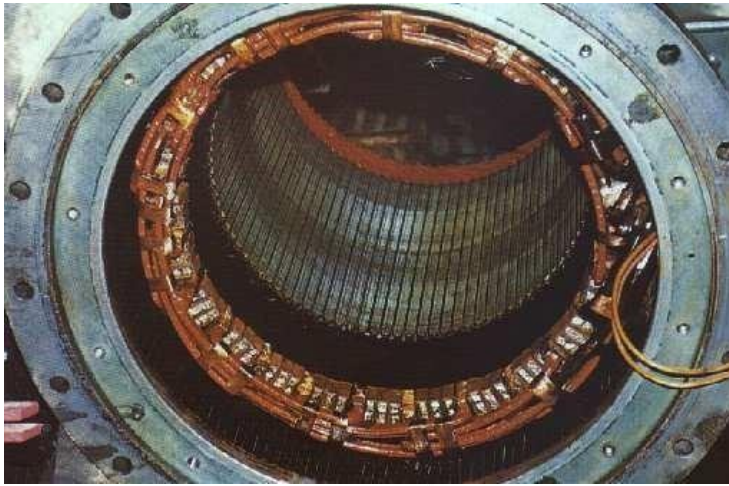
-
- 1. Most hydraulic turbines have to turn at low speeds (between 50 and 300 r/min)**

2. A large number of poles are required on the rotor

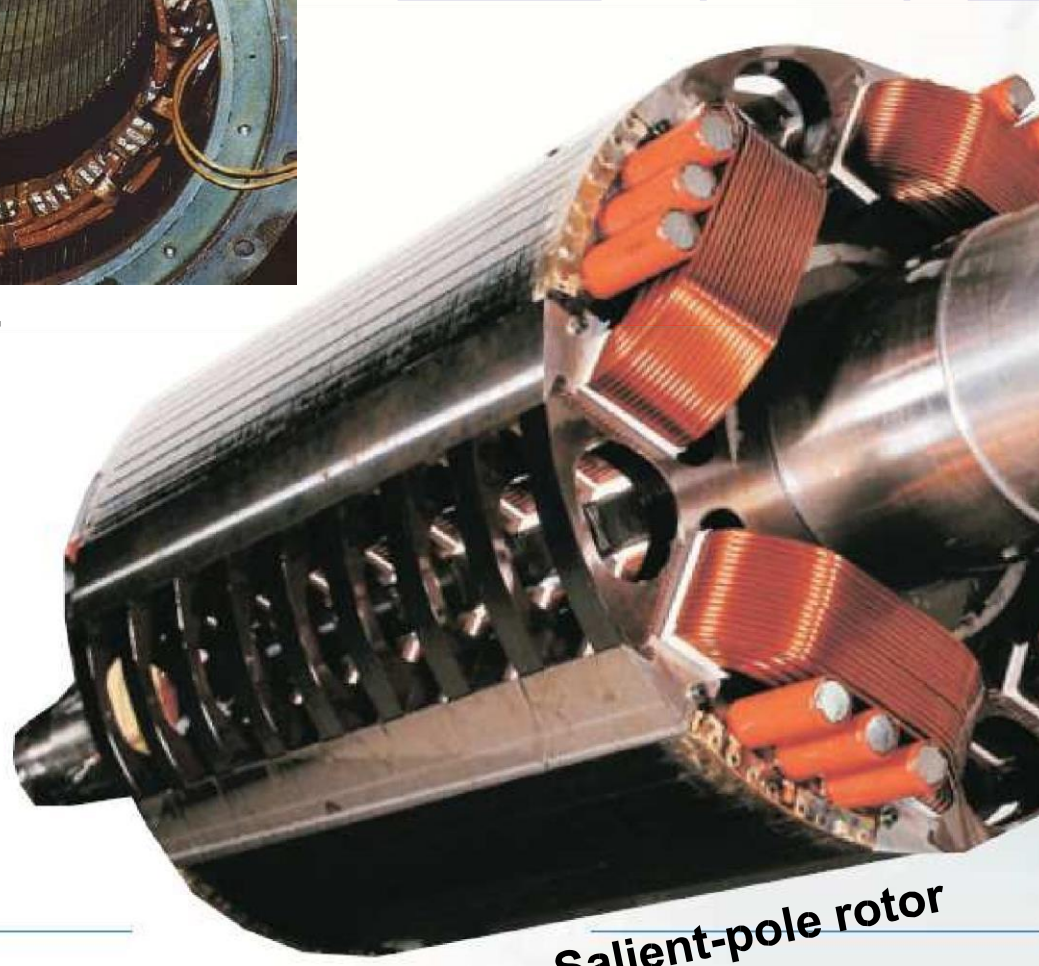


Hydro generator

Salient-Pole Synchronous Generator



Stator



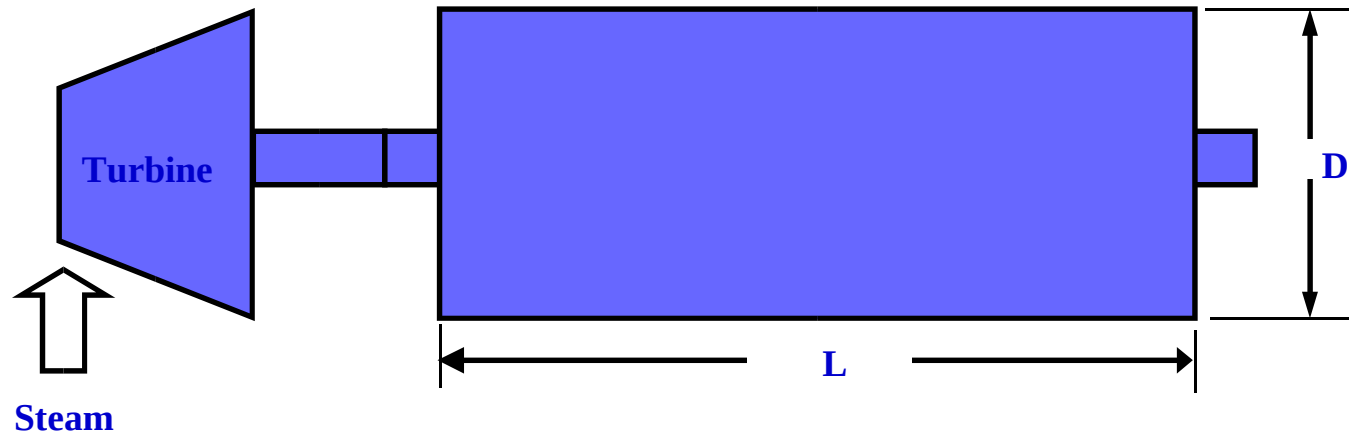
Salient-pole rotor

Cylindrical Type Rotor (Non Salient)

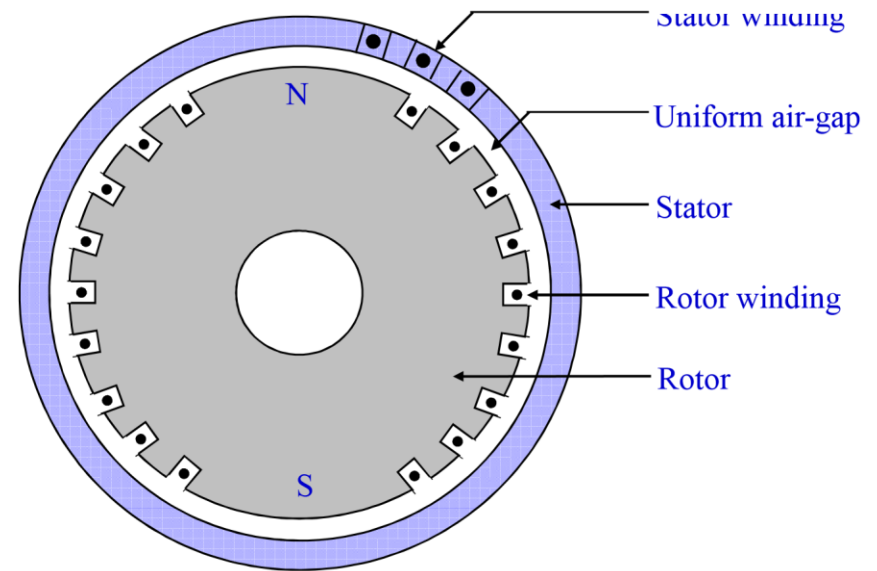
1. The cylindrical rotor has no protruding poles.

2. The rotor of cylindrical rotor generally has two poles.
3. The cylindrical rotor has low diameter (1.2m) and long lengths (2 to 5m).
4. Cylindrical rotors are used for high speed turbo generators, like gas and steam turbine generation system.

Cylindrical-Rotor Synchronous Generator

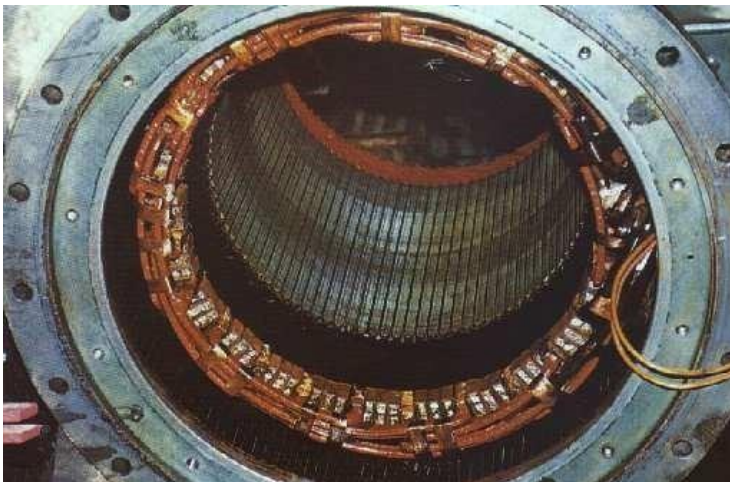


- High speed
- $3600 \text{ r/min} \Rightarrow 2\text{-pole}$
- $1800 \text{ r/min} \Rightarrow 4\text{-pole}$
- Direct-conductor cooling (using hydrogen or water as coolant)
- Rating up to 2000 MVA

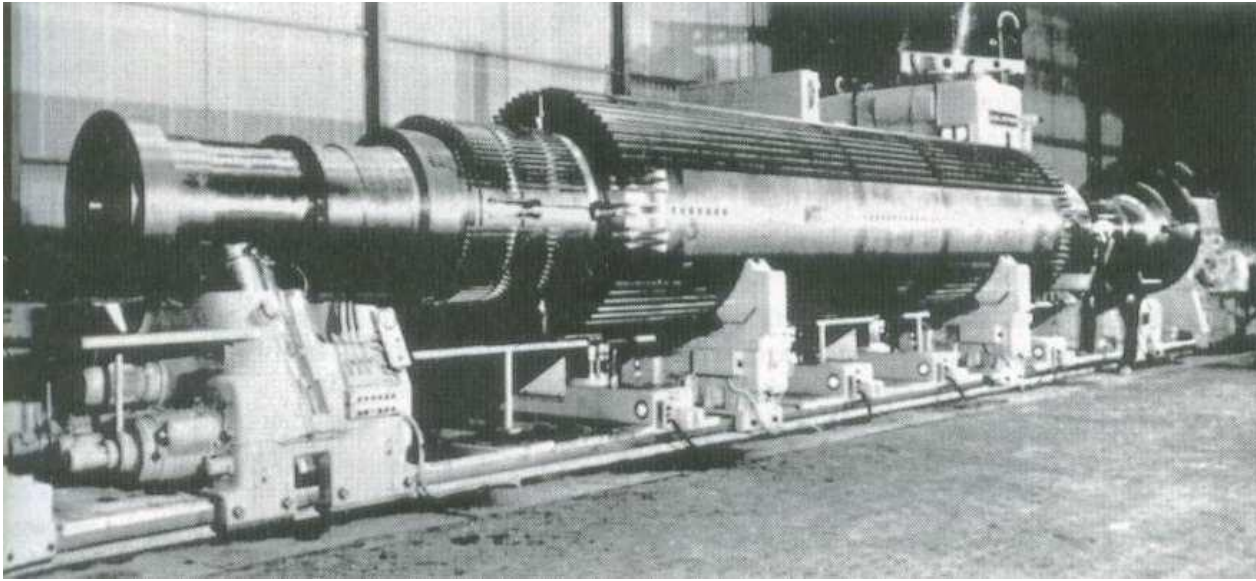


Turbo generator

Cylindrical-Rotor Synchronous Generator



Stator



Cylindrical rotor

Difference Between Salient & Non Salient Rotor

Poles 1. The salient rotor has protruding poles.

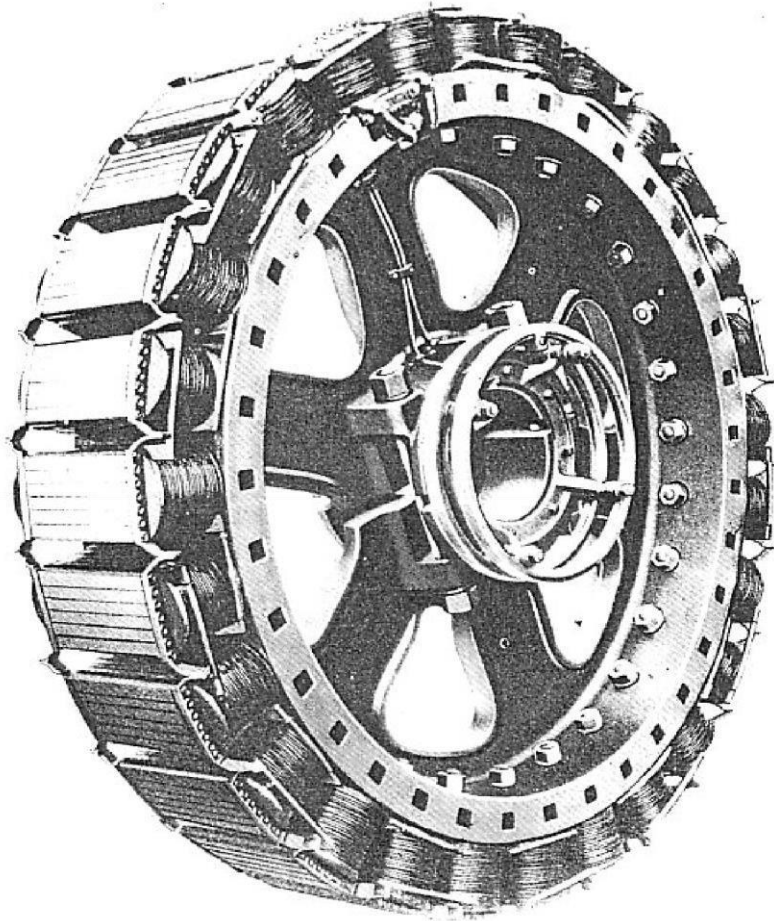
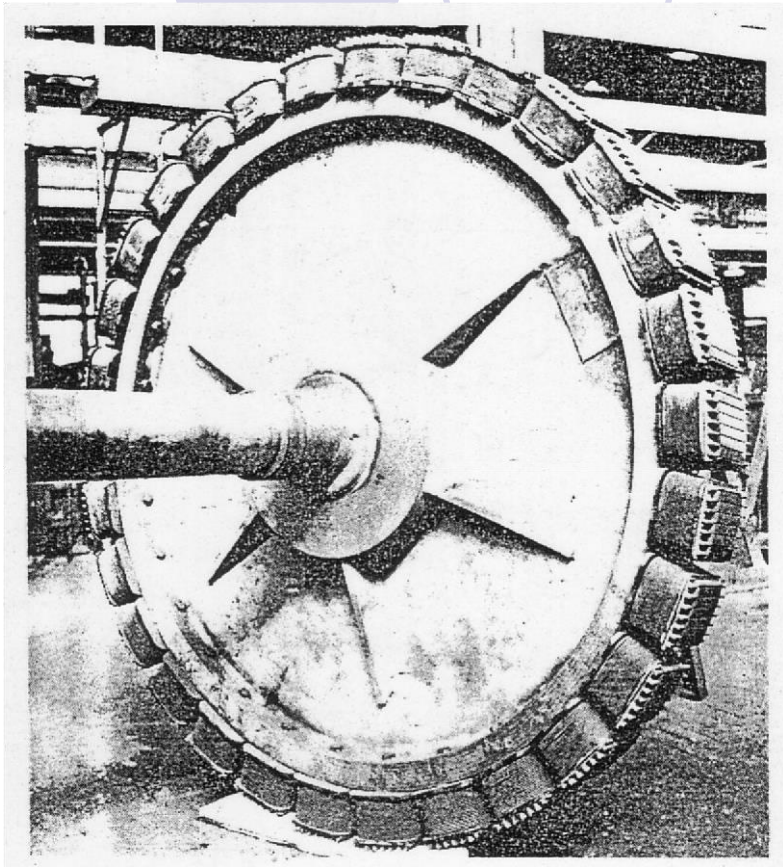
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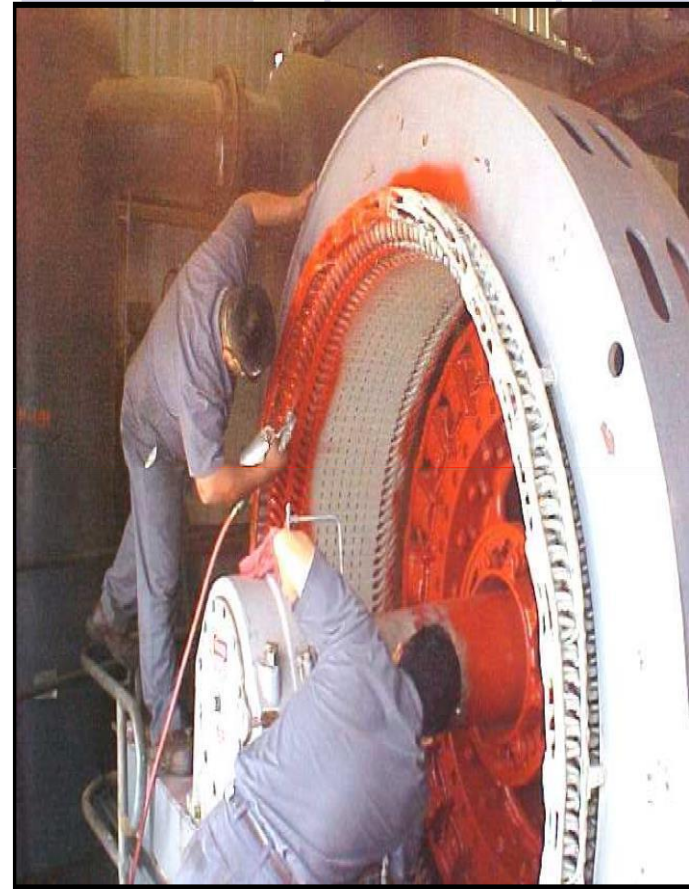
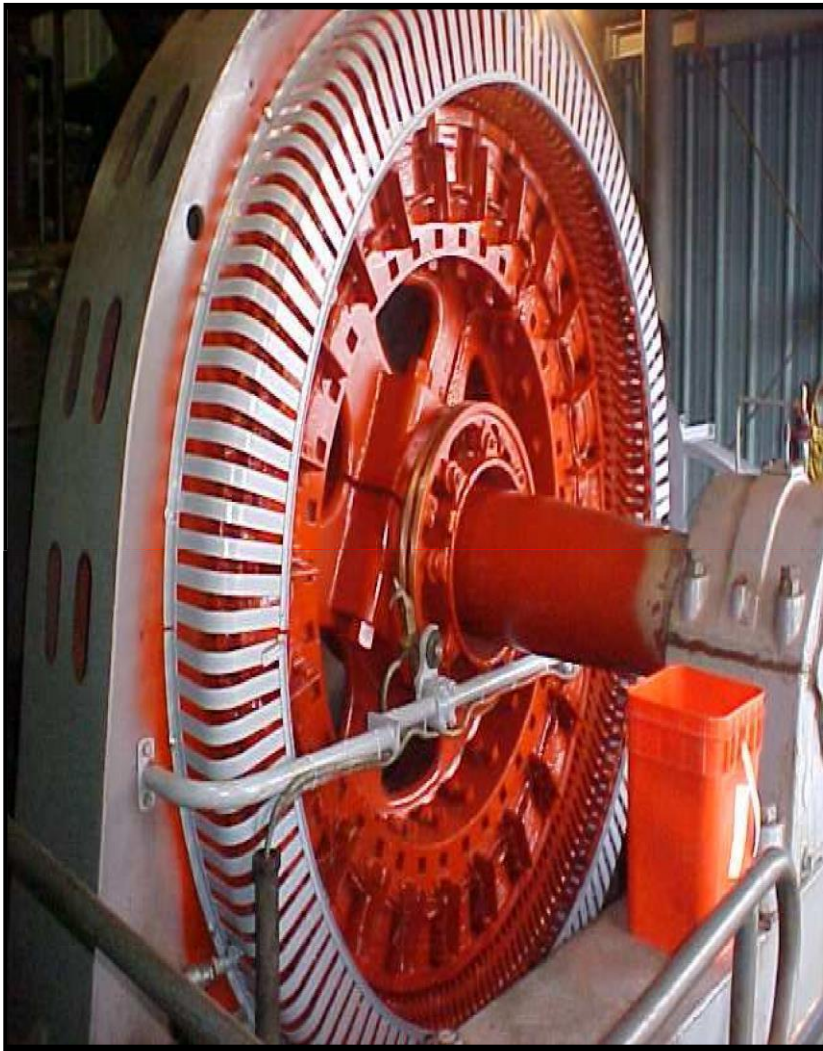
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Salient pole rotor are used for low speed Hydro generators. poles.

3. The cylindrical rotor has low diameter (1.2m) and long lengths (2 to 5m).

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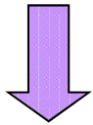




SYNCHRONOUS

Operation Principle

The rotor of the generator is driven by a prime-mover



A dc current is flowing in the rotor winding which produces a rotating magnetic field within the machine



The rotating magnetic field induces a three-phase voltage in the stator winding of the generator

Excitation System



A dc current must be supplied to the field circuit on the rotor. Since the rotor is rotating, a special arrangement is required to get the dc power to its field windings.

SYNCHRONOUS

The common ways are: □ Supply the dc power from an external dc source to the rotor by means of slip rings and brushes. □ Supply the dc power from a special dc power source mounted directly on the shaft of the synchronous generator.

Slip rings & Brush □ Slip rings are metal rings completely encircling the shaft of a machine but insulated from it. One end of the dc rotor winding is tied to each of the 2 slip rings on the shaft of the synchronous machine, and a stationary brush rides on each slip ring.

● A “brush” is a block of graphite like carbon compound that conducts electricity freely but has very low friction, hence it doesn’t wear down the slip ring. If the positive end of a dc voltage source is connected to one brush and the negative end is connected to the other, then the same dc voltage will be applied to the field winding at all times regardless of the angular position or speed of the rotor.

Some problems with slip rings and brushes

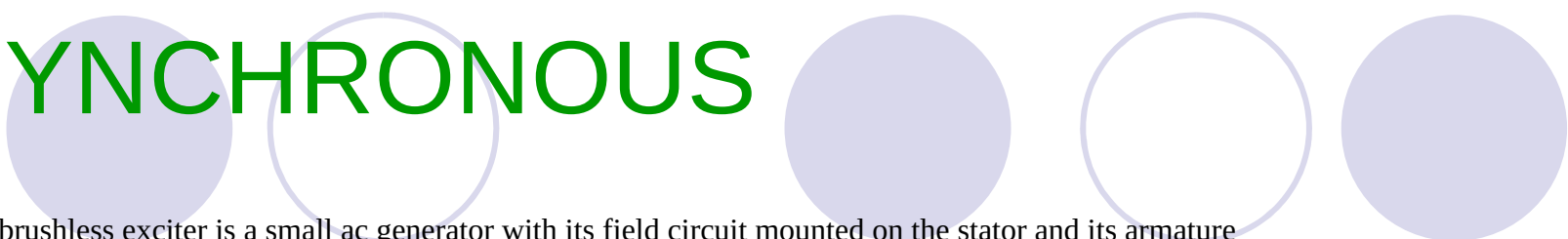
● They increase the amount of maintenance required on the machine, since the brushes must be checked for wear regularly.

● Brush voltage drop can be the cause of significant power losses on machines with larger field currents.

● **Small synchronous machines** – use slip rings and brushes.

● **Larger machines** – brushless exciters are used to supply the dc field current. **Brushless Excitation**

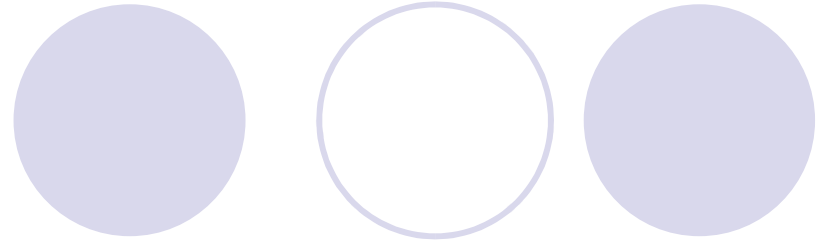
SYNCHRONOUS



- A brushless exciter is a small ac generator with its field circuit mounted on the stator and its armature circuit mounted on the rotor shaft.
- The 3-phase output of the exciter generator is rectified to direct current by a 3-phase rectifier circuit also mounted on the shaft of the generator, and is then fed to the main dc field circuit. By controlling the small dc field current of the exciter generator (located on the stator), we can adjust the field current on the main machine without slip rings and brushes.
- Since no mechanical contacts occur between the rotor and stator, a brushless exciter requires less maintenance.

SYNCHRONOUS

Brushless Excitation



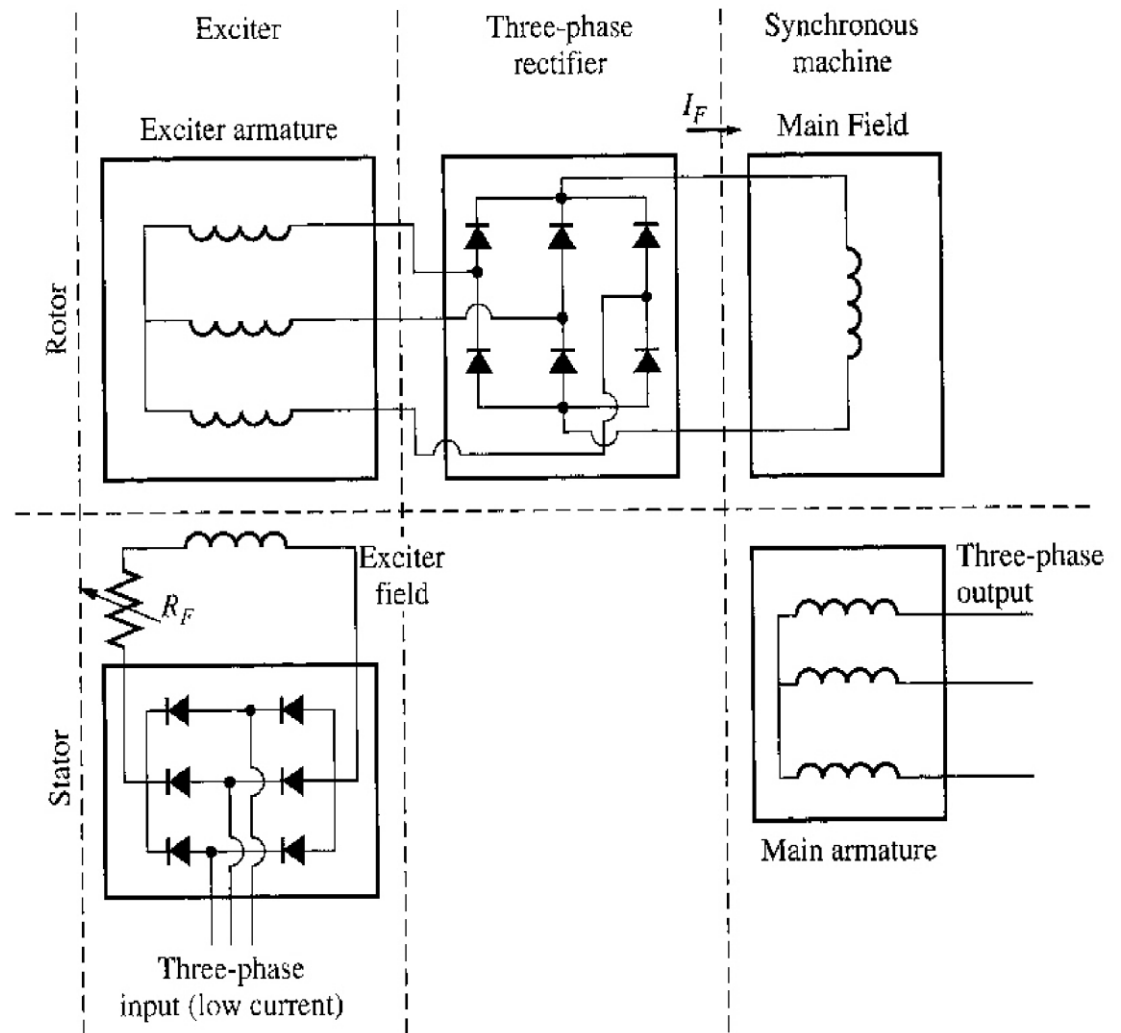
SYNCHRONOUS

Brushless Exciter Circuit:

A small 3-phase current is rectified and used to supply the field circuit of the exciter, which is located on the stator. The output of the armature circuit of the exciter (on the rotor) is then rectified and used to supply the field current of the main machine. Pilot Excitation System □

To make the excitation of a generator completely independent of any external power sources, a small pilot exciter can be used.

- A pilot exciter is a small ac generator with permanent magnets mounted on the rotor shaft and a 3-phase winding on the stator. It produces the power for the field circuit of the exciter, which in turn controls the field circuit of the main machine. If a pilot exciter is included on the generator shaft, then no external electric power is required.



SYNCHRONOUS

Pilot Excitation System Circuit

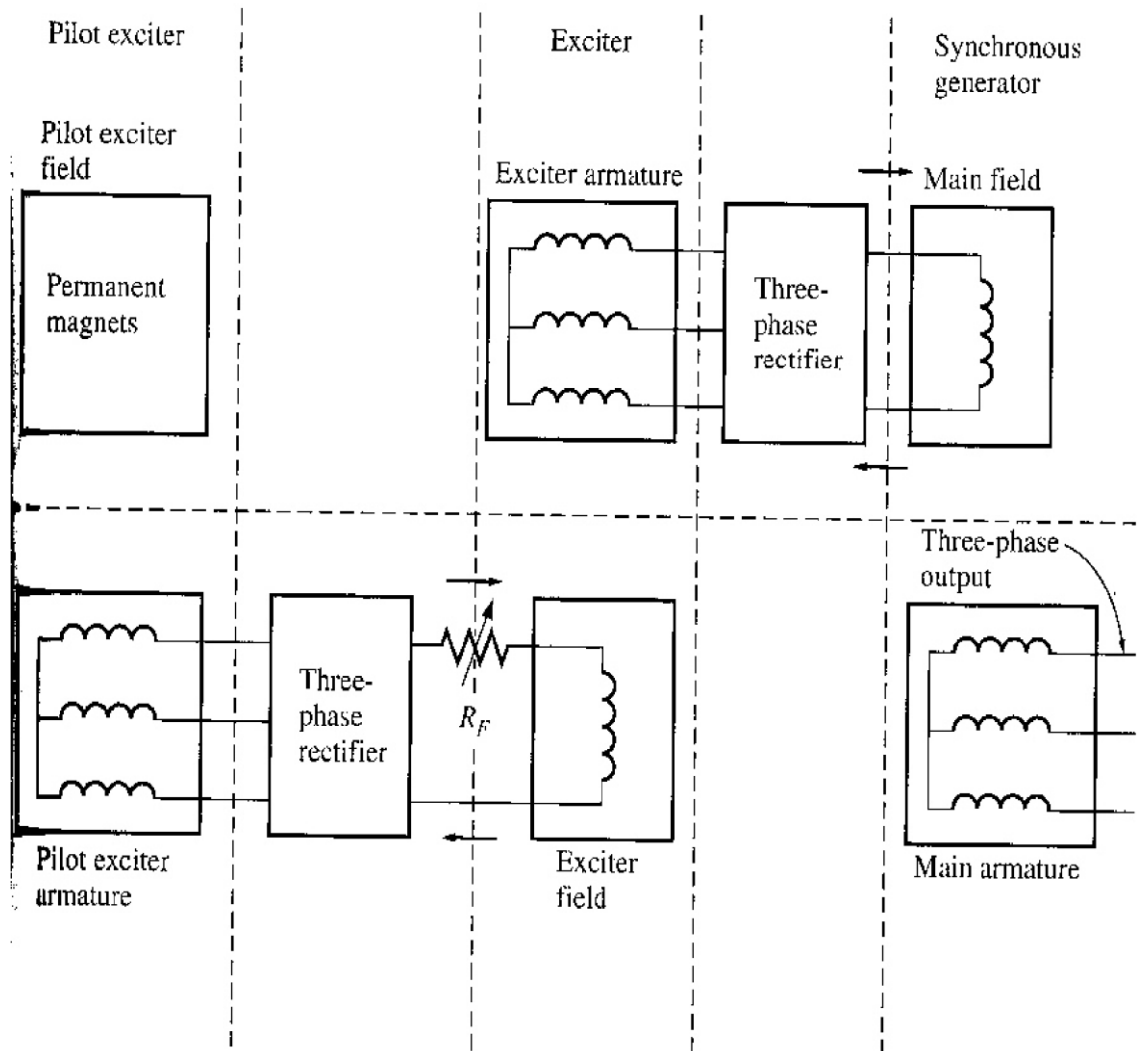
A brushless excitation scheme that includes a pilot exciter. The permanent magnets of the pilot exciter produce the field current of the exciter, which in turn produces the field current of the main machine.

Pilot Excitation System

Even though machines with brushless exciters do not need slip rings and brushes, they still include the slip rings and brushes so that an auxiliary source of dc field current is available in emergencies.

The Speed of Rotation of a Synchronous Generator

Synchronous generators are by definition *synchronous*, meaning that the electrical frequency produced is



SYNCHRONOUS

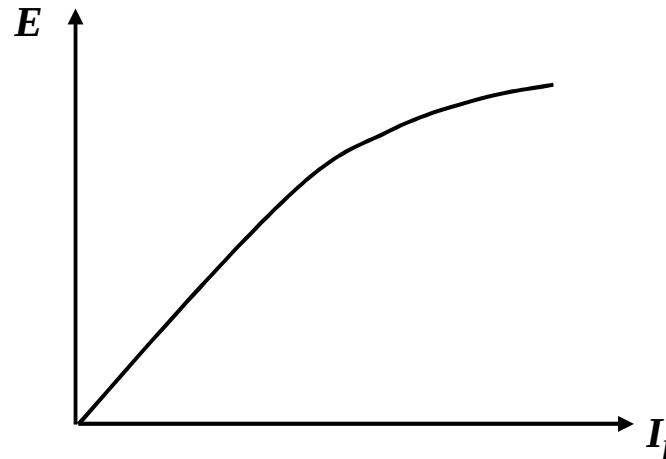
locked in or synchronized with the mechanical rate of rotation of the generator. A synchronous generator's rotor consists

of an electromagnet to which direct current is supplied. The rotor's magnetic field points in the direction the rotor are turned. Hence, the rate of rotation of the magnetic field in the machine is related to the stator electrical

The generated voltage of a synchronous generator is given by

$$E = K_c \phi f_e$$

where ϕ = flux in the machine (function of I_f) f_e = electrical frequency K_c = synchronous machine constant



Saturation characteristic of a synchronous generator.

SYNCHRONOUS

Voltage Regulation

A convenient way to compare the voltage behaviour of two generators is by their *voltage regulation* (VR).

The VR of a synchronous generator at a given load, power factor, and at rated speed is defined as

$$VR = \frac{E_{nl} - V_{fl}}{V_{fl}} \times 100\%$$

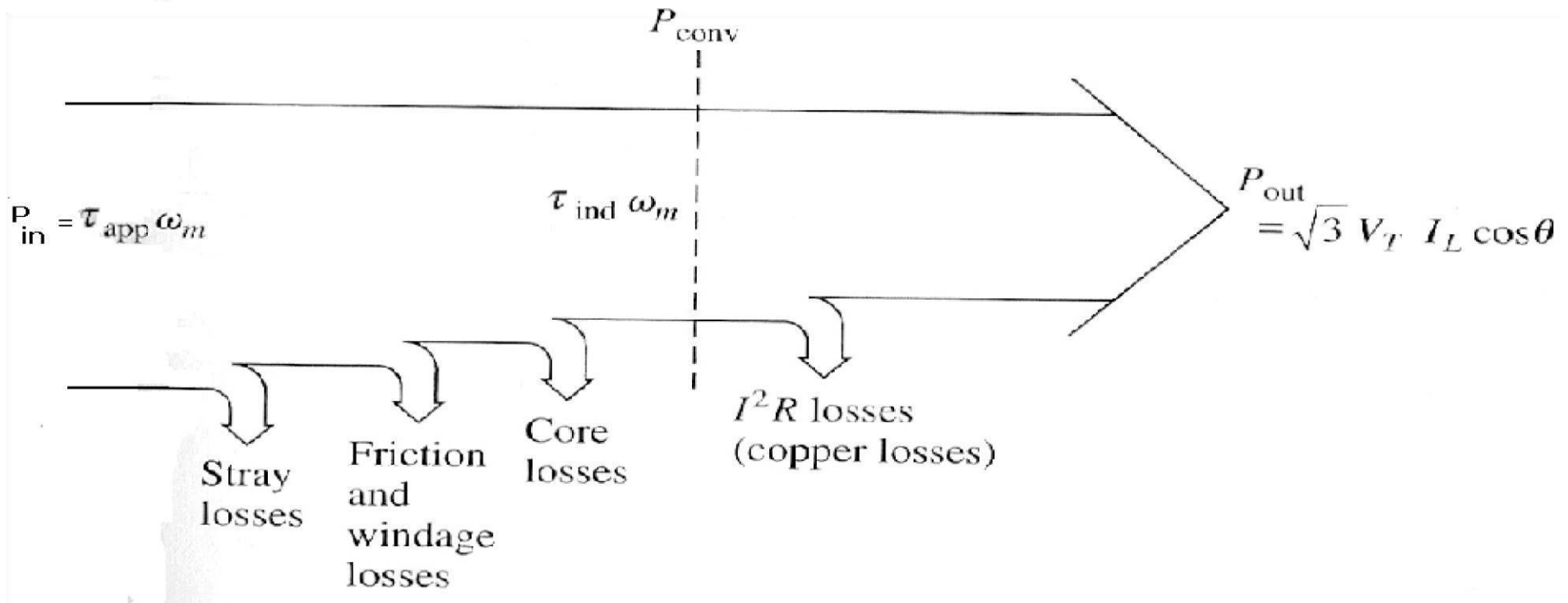
Where V_{fl} is the full-load terminal voltage, and E_{nl} is the no-load terminal voltage (internal voltage) at rated speed when the load is removed without changing the field current.

For lagging power factor (PF), VR is fairly positive, for unity PF, VR is small positive and for leading PF, VR is negative.

SYNCHRONOUS GENERATORS

Power and Torque | Not all mechanical power going to a synchronous generator becomes electrical power the difference between input & output power represent losses, Power flow Diagramè

SYNCHRONOUS



SYNCHRONOUS GENERATORS

P

Power and Torque Input mechanical power is shaft power in generator $P_{in} = T_{prime-mover} \omega_m$

($T_{prime-mover} \equiv T_{app}$) $P_{conv} = T_{generated} \omega_m = 3EA I_A \cos \gamma$

(γ angle between EA and IA)

SYNCHRONOUS

($T_{\text{generated}} \equiv T_{\text{ind}}$)

Difference between P_{in} and P_{conv}
in generator represents
mechanical, core, and stray losses
of machine

GENERATORS

Power and Torque

Real elect. output power of syn. Gen. in line quantities: $P_{\text{out}} = \sqrt{3} V_T I_L \cos\theta$

reactive power output:

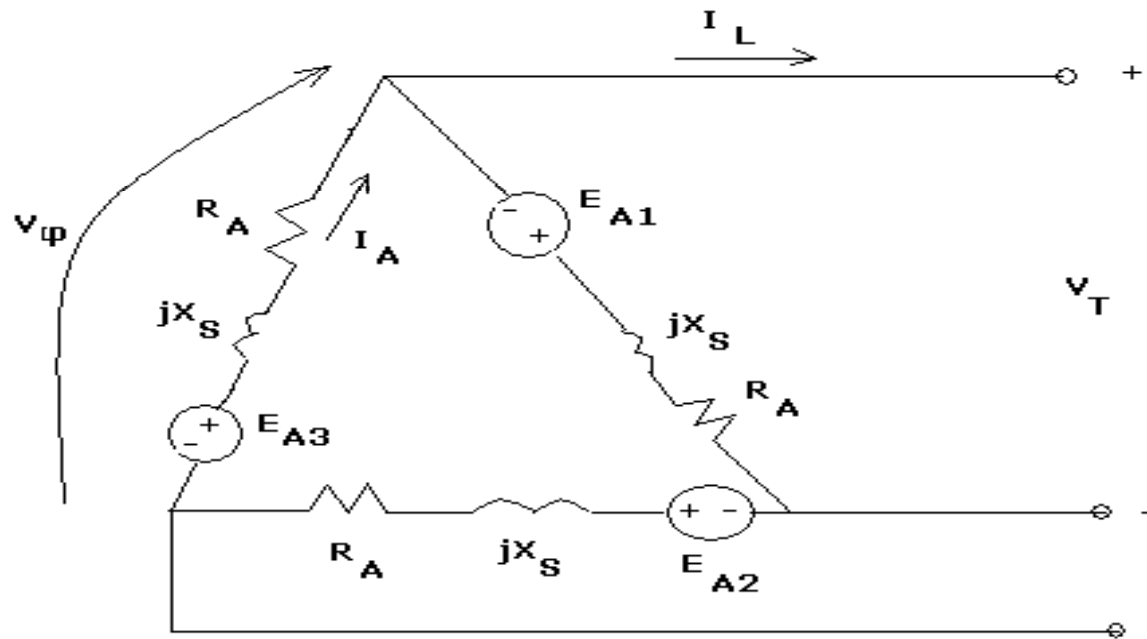
$$Q_{\text{out}} = \sqrt{3} V_T I_L \sin\theta$$

$$Q_{\text{out}} = \sqrt{3} V_{\phi} I_A \sin\theta$$

ignoring armature resistance R_A ($X_S \gg R_A$), a useful relation can be derived to approximate output power of Gen.

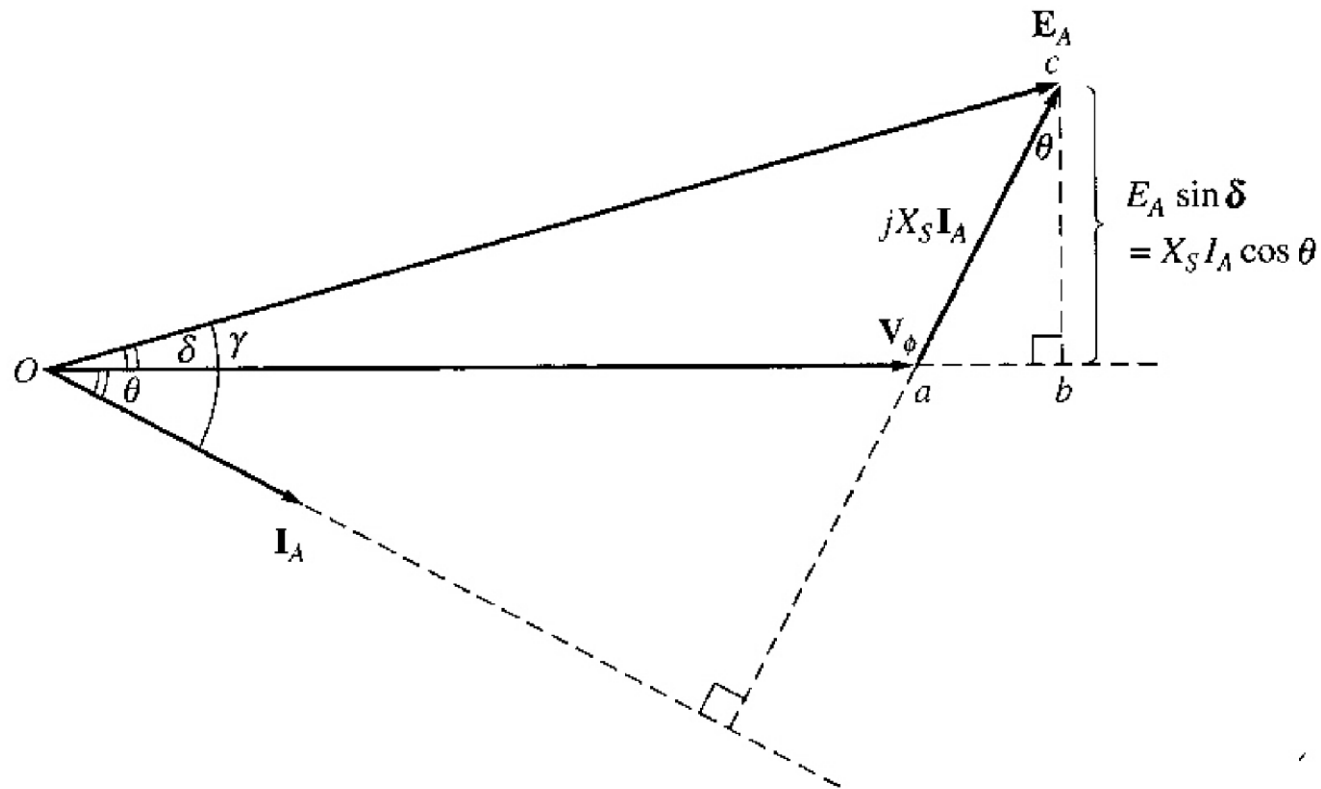
SYNCHRONOUS

Generator Equivalent circuit for Δ connection



gram employed

SYNCHRONOUS



GENERATORS phasor diagram

The Equivalent Circuit of a Synchronous Generator

- The internal voltage E_f produced in a machine is not usually the voltage that appears at the terminals of the generator. o The only time E_f is same as the output voltage of a phase is when there is no armature current flowing in the machine.
- There are a number of factors that cause the difference between E_f and V_t :
 - The distortion of the air-gap magnetic field by the current flowing in the stator, called the armature reaction □
 - The self-inductance of the armature coils.
 - The resistance of the armature coils.
 - The effect of salient-pole rotor shapes.

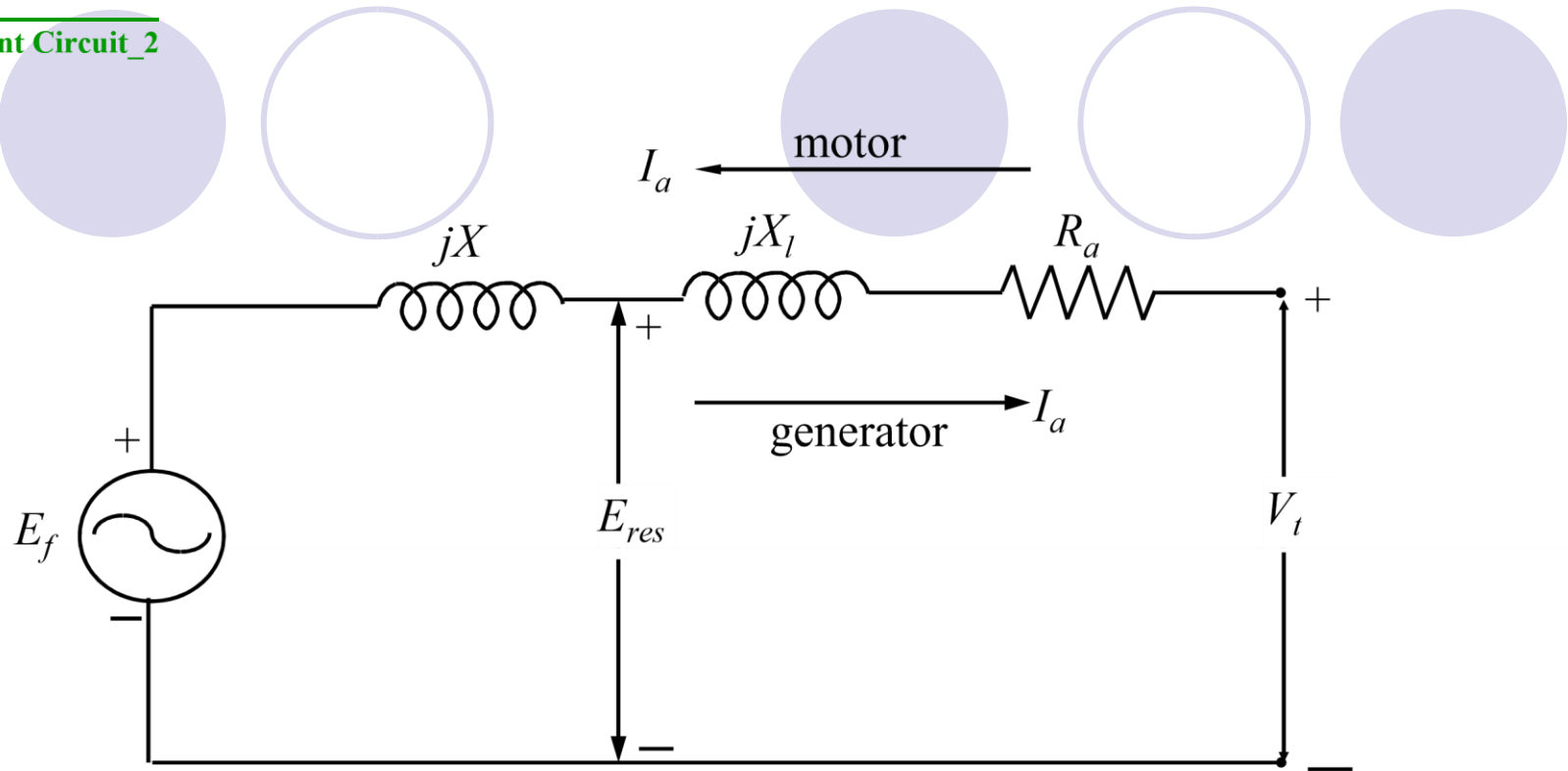
The Equivalent Circuit of a Synchronous Generator

Armature Reaction

When the rotor is spun, a voltage E_A is induced in the stator windings. If a load is attached to the terminals of the generator, a current flows. But a 3-phase stator current flow will produce a magnetic field of its own. This stator magnetic field will distort the original rotor magnetic field, changing the resulting phase voltage.

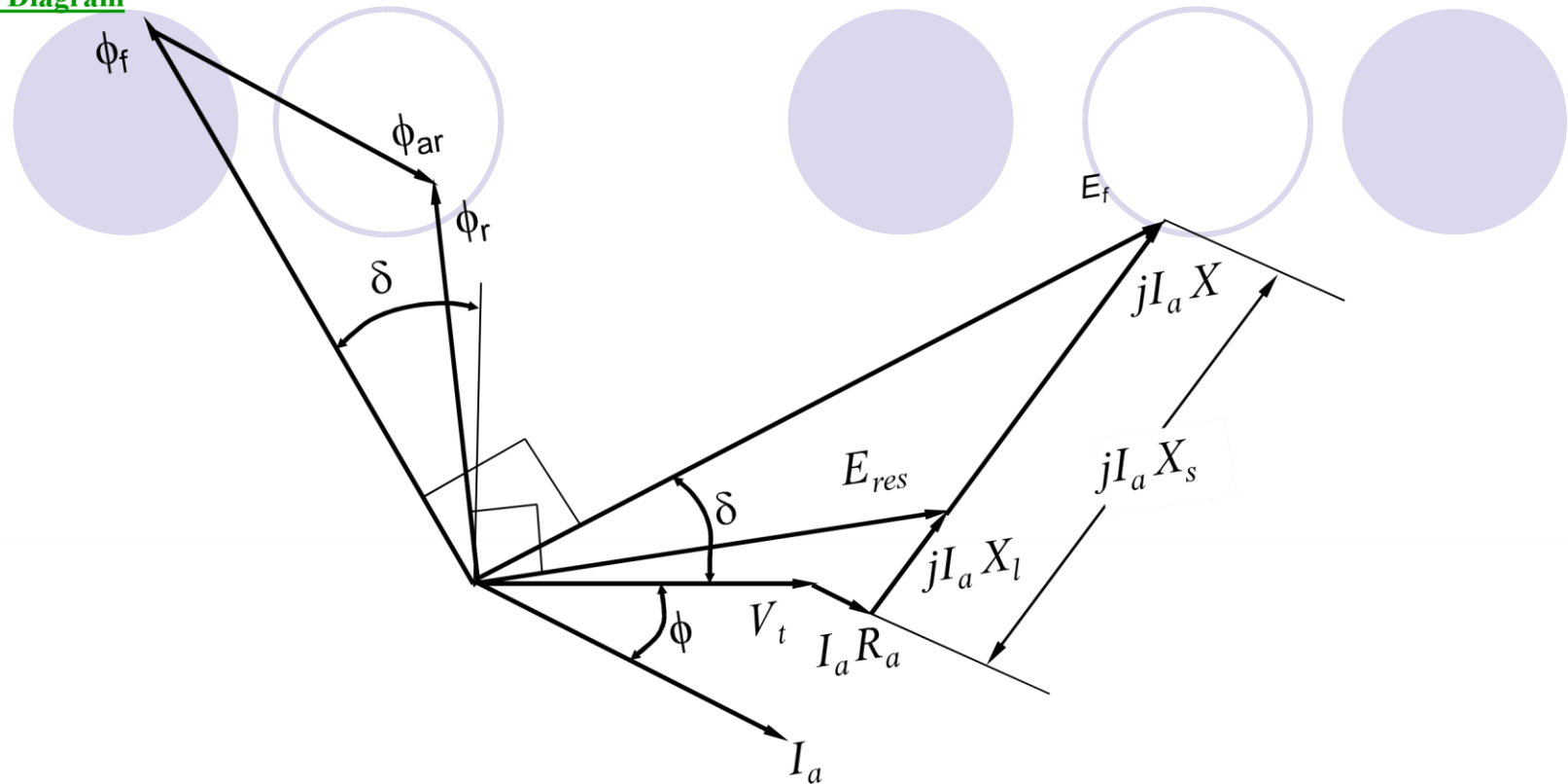
This effect is called armature reaction because the armature (stator) current affects the magnetic field, which produced it in the first place.

Equivalent Circuit_2



Equivalent circuit of a cylindrical-rotor synchronous machine

Phasor Diagram

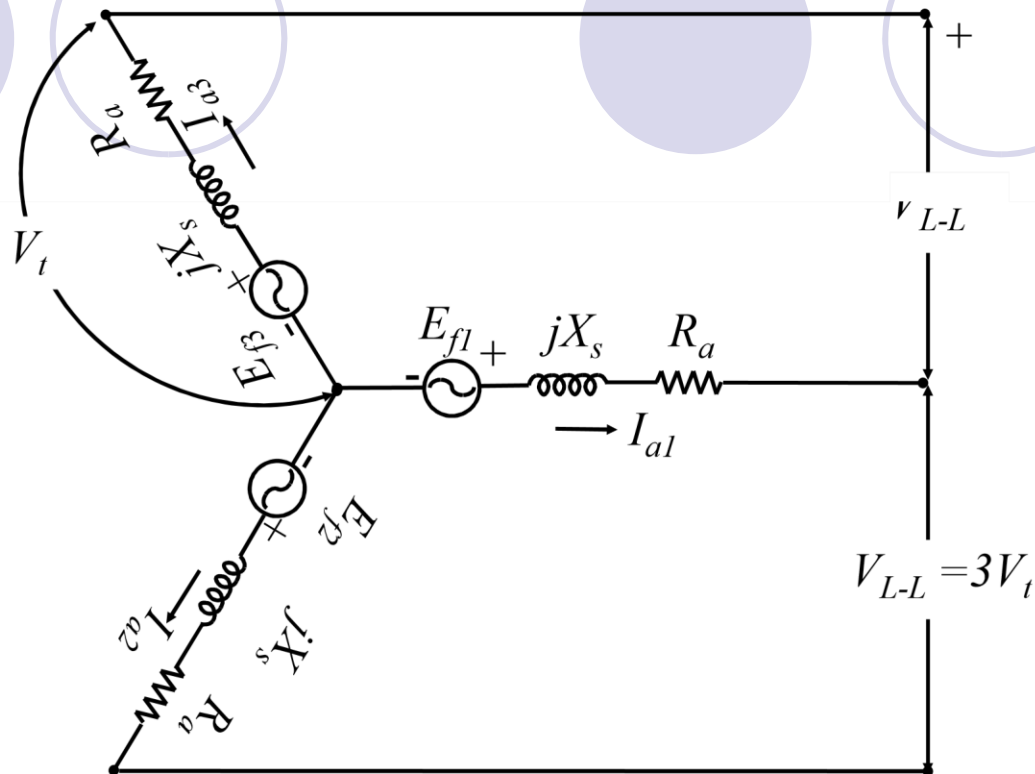


Phasor diagram of a cylindrical-rotor synchronous generator,
for the case of lagging power factor

Lagging PF: $|V_t| < |E_f|$ for overexcited condition
Leading PF: $|V_t| > |E_f|$ for underexcited condition

Three-phase equivalent circuit of a cylindrical-rotor synchronous machine

The voltages and currents of the three phases are 120° apart in angle, but otherwise the three phases are identical.



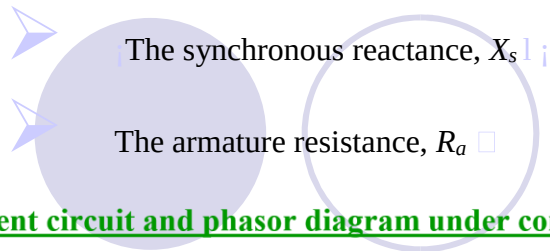
Determination of the parameters of the equivalent circuit from test data

The equivalent circuit of a synchronous generator that has been derived contains three quantities that must be determined in order to completely describe the behaviour of a real synchronous generator:

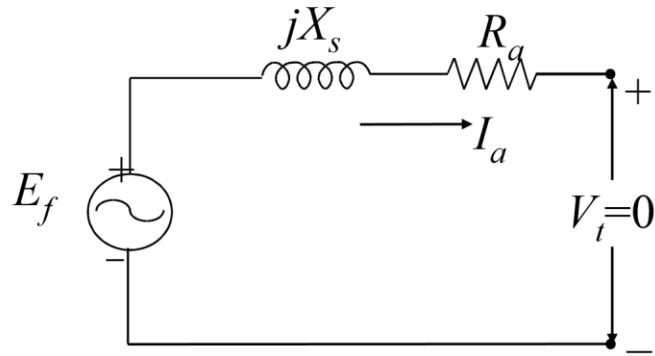
i



The saturation characteristic: relationship between I_f and ϕ (and therefore between I_f and E_f)

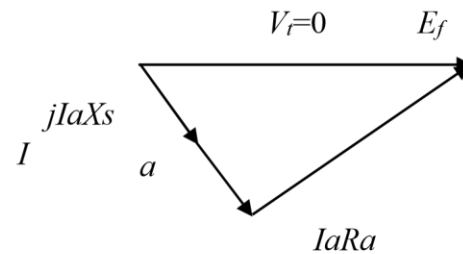


Equivalent circuit and phasor diagram under condition

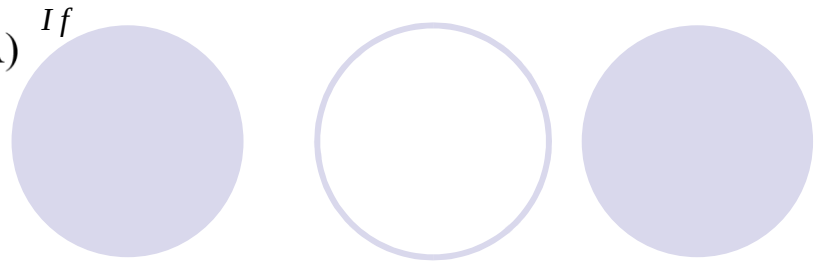
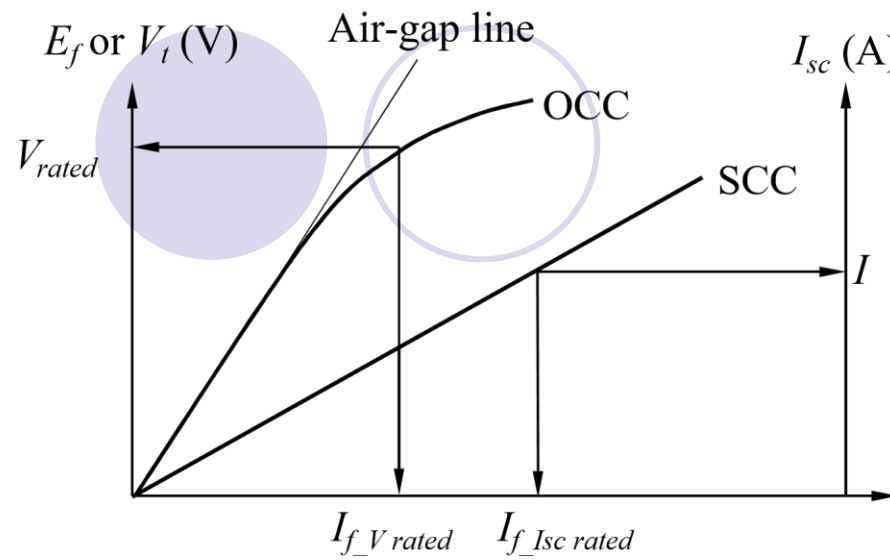


Short-circuit

Ratio



Another parameter used to describe synchronous generators is the short-circuit ratio (SCR). The SCR of a generator defined as the ratio of the *field current required for the rated voltage at open circuit* to the *field current required for the rated armature current at short circuit*. SCR is just the reciprocal of the per unit value of the saturated synchronous reactance calculated by



Operation of AC Generators in Parallel with Large Power Systems

- Isolated synchronous generator supplying its own load is very rare (emergency generators).
- In general applications more than one generator operating in parallel to supply loads.
- In Pakistani national grid hundreds of generators share the load on the system.

Parallel operation of synchronous generators

There are several major advantages to operate generators in parallel:

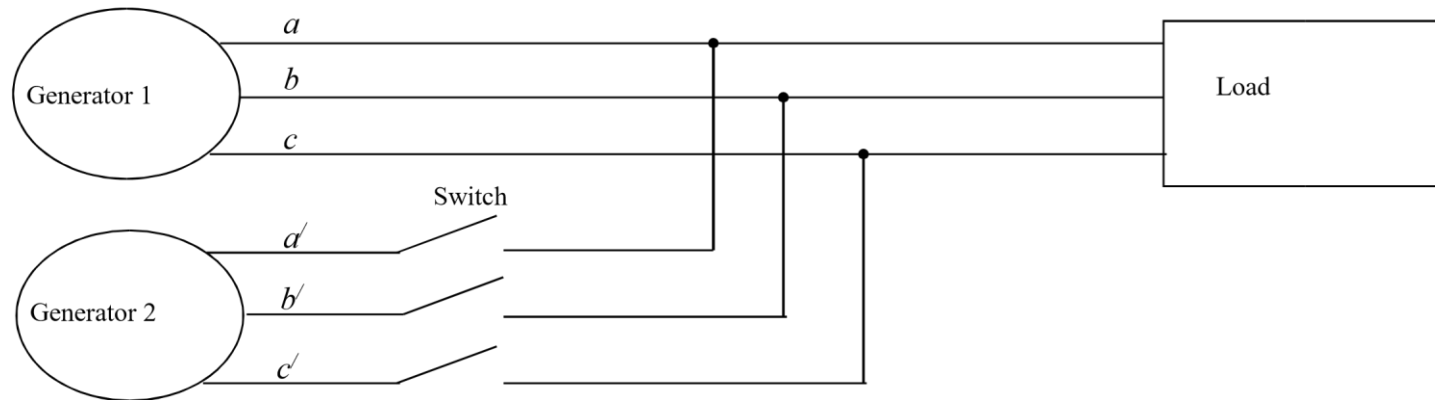
- Several generators can supply a bigger load than one machine by itself.
- Having many generators increases the reliability of the power system.
- It allows one or more generators to be removed for shutdown or preventive maintenance.
- ☐ If only one generator employed &

not operating near full load, it will be relatively inefficient

Synchronization

Before connecting a generator in parallel with another generator, it must be synchronized. A generator is said to be synchronized when it meets all the following conditions:

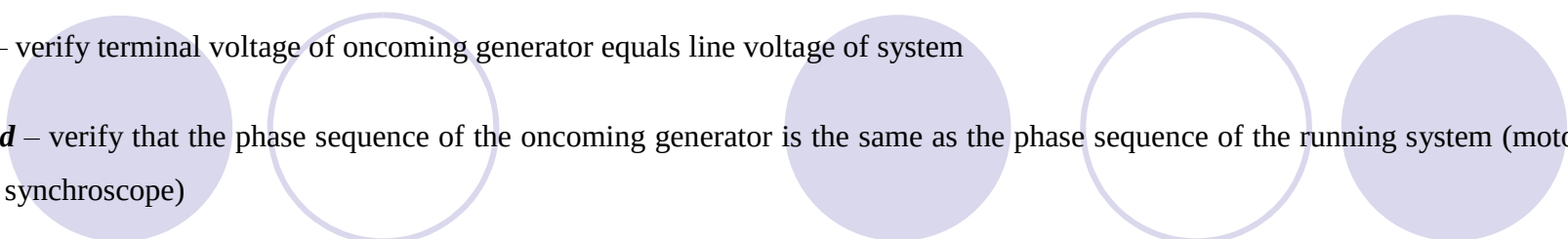
- The *rms line voltages* of the two generators must be equal. | The two generators must have the same *phase sequence*.
- The *phase angles* of the two *a* phases must be equal.
- The *oncoming generator frequency* is equal to the running system frequency.



Concept of the infinite bus

When a synchronous generator is connected to a power system, the power system is often so large that nothing the operator of the generator does will have much of an effect on the power system. An example of this situation is the connection of a single generator to the Pakistan power grid. Our Pakistani power grid is so large that no reasonable action on the part of one generator can cause an observable change in overall grid frequency. This idea is idealized in the concept of an infinite bus. *An infinite bus is a power system so large that its voltage and frequency do not vary regardless of how much real or reactive power is drawn from or supplied to it.*

Procedure

- 
- **First** – verify terminal voltage of oncoming generator equals line voltage of system
 - **Second** – verify that the phase sequence of the oncoming generator is the same as the phase sequence of the running system (motor, bulbs, synchroscope)
 - **Third** – adjust the frequency of the oncoming unit to be slightly higher than the frequency of the running system. (When a synchronous Generator connected to a power system:

1-The real power versus frequency characteristic of such a system

2-And the reactive power voltage characteristic

Operation of AC Generators in Parallel with Large Power Systems

Behavior of a generator connected to a large system.

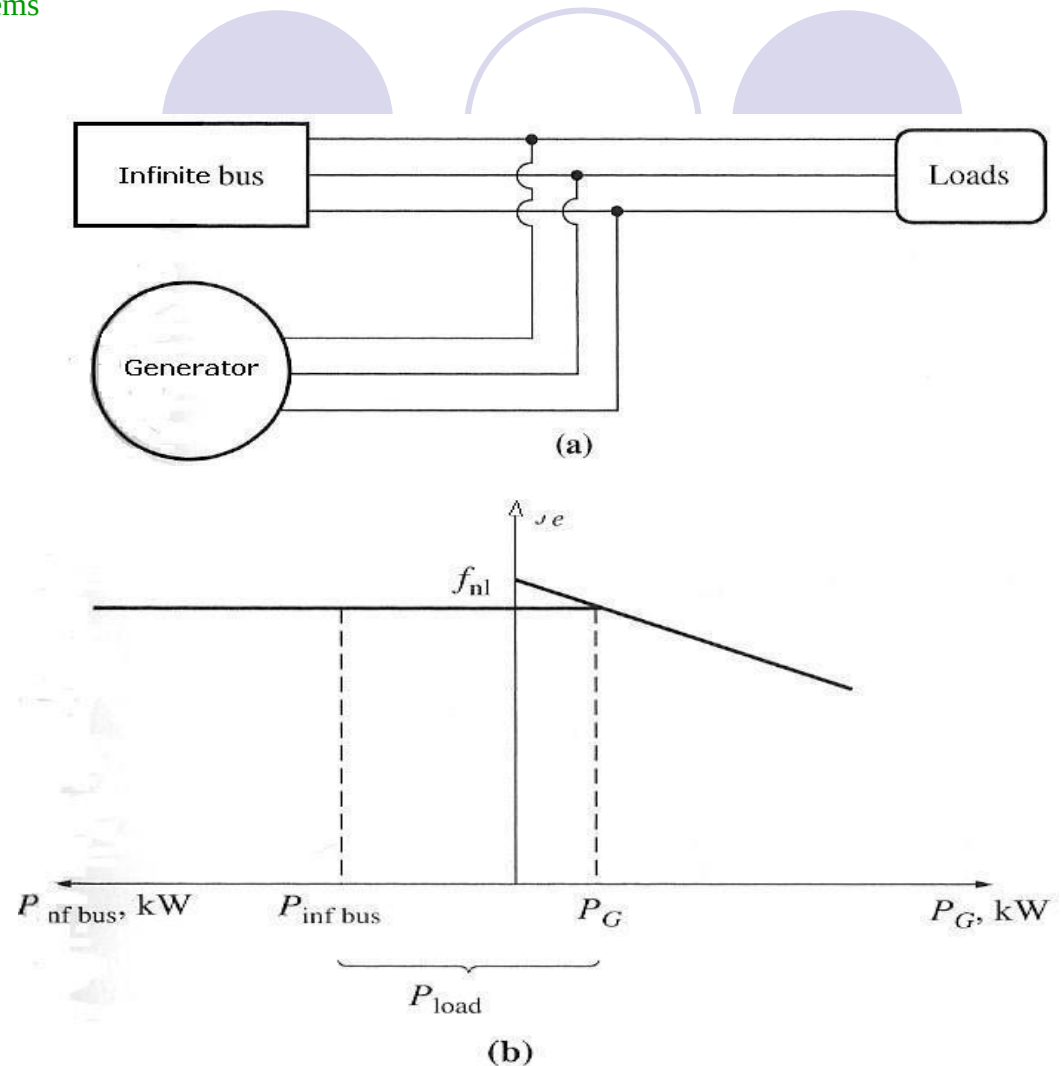
A generator connected in parallel with a large system as shown: □ Frequency & voltage of all machines must be the same, their real power frequency (& reactive power-voltage) characteristics plotted back to back → with Large

Power Systems Assume generator just been paralleled with infinite bus, generator will be “floating” on the line, supplying a small amount of real power and little or no reactive power. Suppose generator paralleled, however its frequency being slightly lower than system’s operating frequency →

At this frequency power supplied by generator is less than system’s operating frequency, generator will consume energy and runs as motor . with Large Power Systems

In order that a generator comes on line and supply power instead of consuming it, we should ensure that oncoming machine’s frequency is adjusted higher than running system’s frequency.

Many generators have “reverse-power trip” system. And if such a generator ever starts to consume power it will be automatically disconnected from line.



Active and reactive power-angle characteristics

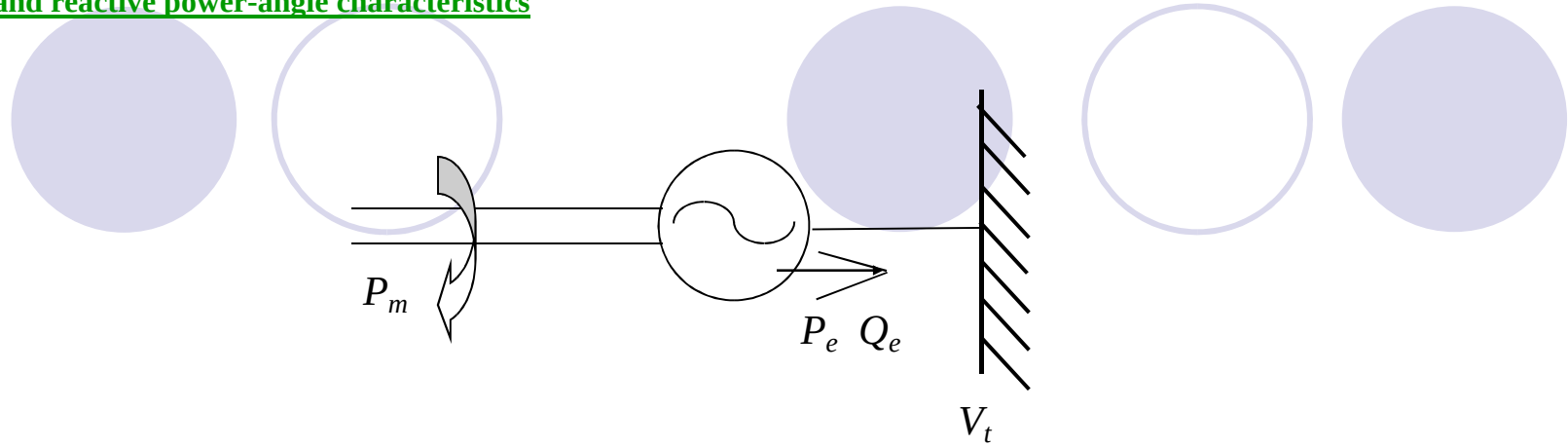


Fig. Synchronous generator connected to an infinite bus.

- $P > 0$: generator operation | $P < 0$: motor operation
- Positive Q : delivering inductive vars for a generator action or receiving inductive vars for a motor action
- Negative Q : delivering capacitive vars for a generator action or receiving capacitive vars for a motor action

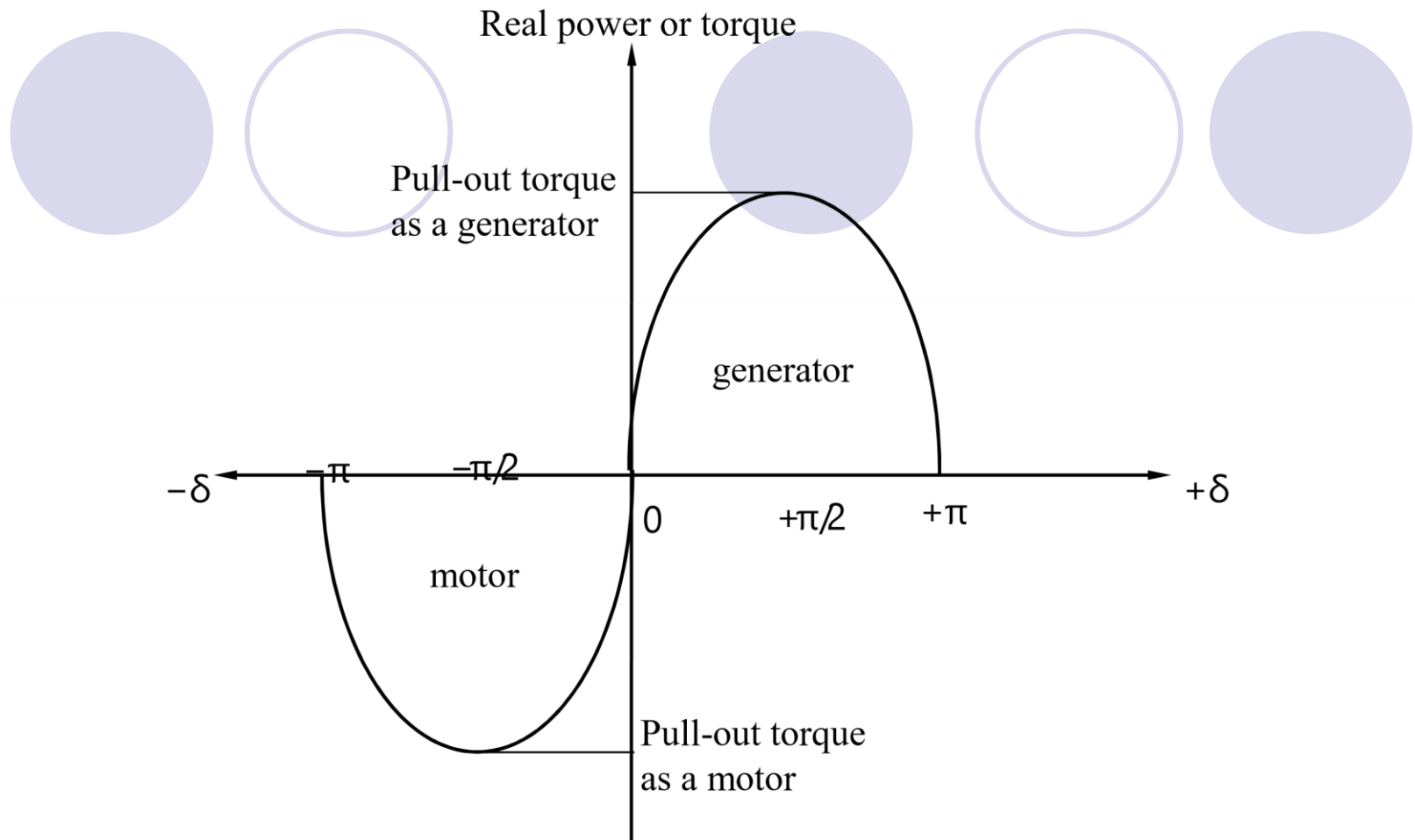
The real and reactive power delivered by a synchronous generator or consumed by a synchronous motor can be expressed in terms of the terminal voltage V_t , generated voltage E_f , synchronous impedance Z_s , and the power angle or torque angle δ .

Active and reactive power-angle characteristics

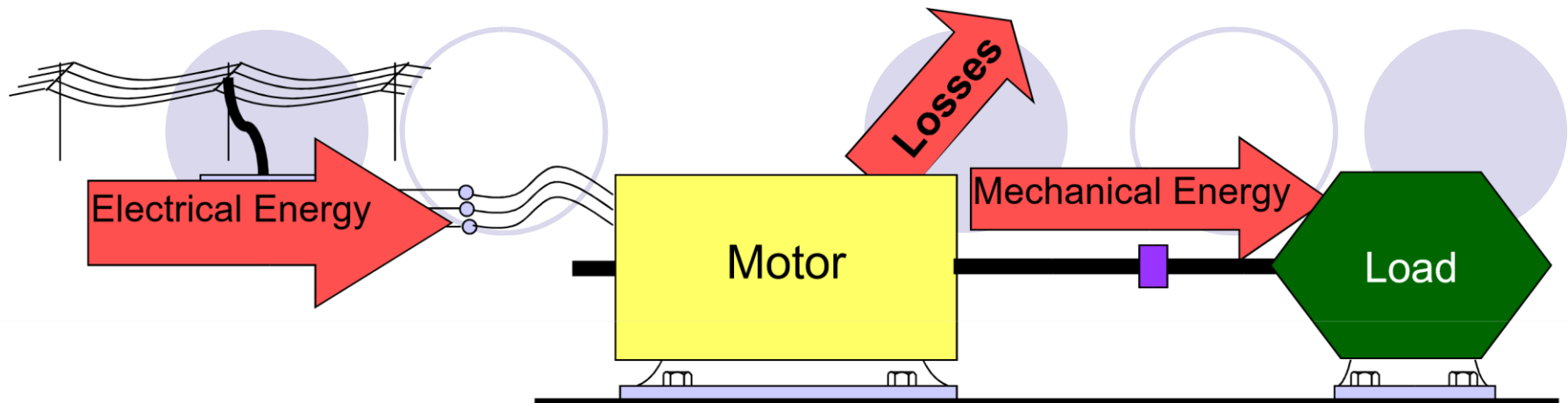
$S = P + jQ = V_t I_a^*$ where:

V_t = terminal voltage per phase

I_a^* = complex conjugate of the armature current per phase



Steady-state stability limit

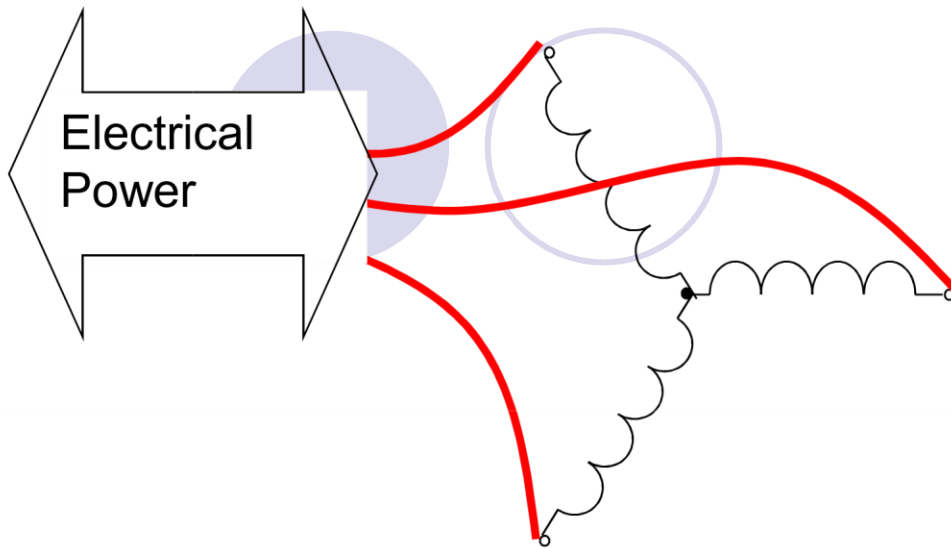


Motor

$$P_{in} = P_{out} + \text{Losses}$$

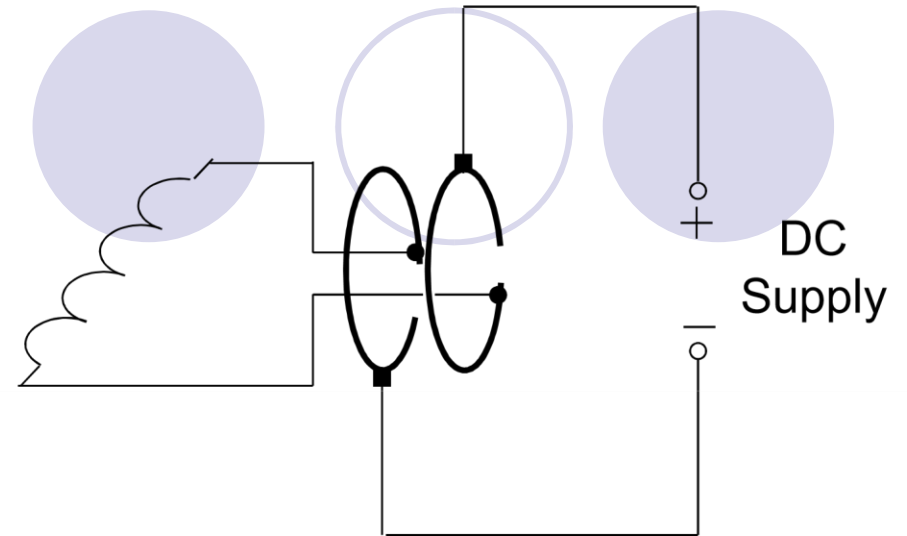
Motor:

$$\text{Eff}\% = \frac{P_{out}}{P_{in}} \times 100$$



Stator

- Identically wound to an induction motor.
- Connected to supply.



Rotor

- Constant DC field
- Connected to supply via slip rings.

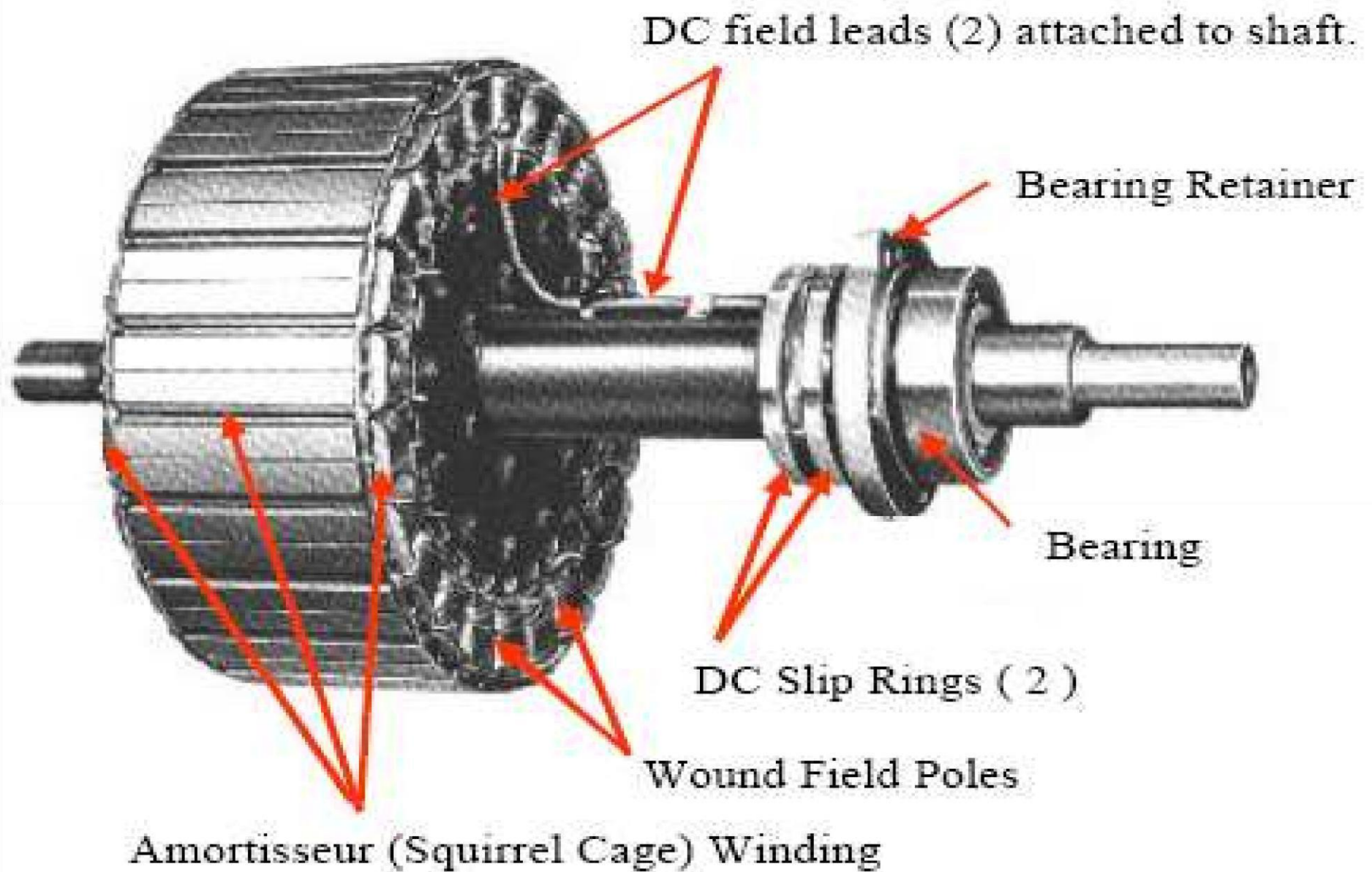
Operation Principle

- The field current of a synchronous motor produces a steady-state magnetic field B_R . A three-phase set of voltages is applied to the stator windings of the motor, which produces a three-phase current flow in the windings. This three-phase set of currents in the armature winding produces a uniform rotating magnetic field of B_s .
- Therefore, there are two magnetic fields present in the machine, and *the rotor field will tend to line up with the stator field*, just as two bar magnets will tend to line up if placed near each other.
- Since the stator magnetic field is rotating, the rotor magnetic field (and the rotor itself) will try to catch up.
- The larger the angle between the two magnetic fields (up to a certain maximum), the greater the torque on the rotor of the machine.

Synchronous Motor Rotors

The Salient-Pole unit shown at the right is a brush-type rotor that uses slip rings for application of the DC field current.

- Low voltage DC is used for the rotating field. 120 VDC and 250 VDC are typical.
- Slip ring polarity is not critical and should be periodically reversed to equalize the wear on the slip rings. The negative polarity ring will sustain more wear than the positive ring due to electrolysis.
- Slip rings are usually made of steel for extended life.



Electric Machinery Photo

why synchronous motor has zero starting ?

A synchronous motor is not self starting. when three phase current displaced in time by 120 degree is applied to the three phase stator windings displaced in space by 120 degree then a rotating magnetic field is produced in the air gap. The rotor tends to align with this rotating field, the rotor windings are supplied by dc .

So constant poles are produced on the rotor. Now suppose the instant when the rotating stator field has its north pole in line with the south pole of the rotor, the rotor will be attracted and tend to move with the stator. but due to its inertia before it starts to move with the stator the south pole of the stator comes in line with the rotor south pole, so now the rotor gets repelling force and it tends to move in the opposite direction as that of the stator. but again before it starts to move the stator north pole comes in line with the rotor south pole. thus overall the effect is that the syn. motor is not self starting. once it is brought to the syn speed by an external prime mover it rotates at syn. speed irrespective of the load.

Synchronous Motors □ Synchronous machines employed to convert electric energy to mechanical energy

● To present the principles of Synchronous motor, a 2-pole synchronous motor considered

● It has the same basic speed, power, & torque equations as Syn. Gen.

$$\tau_{ind} = k B_R \times B_S = \text{counterclockwise}$$

Application of Synchronous Motors

Synchronous motors are usually used in large sizes because in small sizes they are costlier as compared with induction machines.

The principal advantages of using synchronous machine are as follows:

- Power factor of synchronous machine can be controlled very easily by controlling the field current.
- It has very high operating efficiency and constant speed.

- For operating speed less than about 500 rpm and for high-power requirements (above 600KW) synchronous motor is cheaper than induction motor.

In view of these advantages, synchronous motors are preferred for driving the loads requiring high power at low speed; e.g; reciprocating pumps and compressor, crushers, rolling mills, pulp grinders etc.
